Similar questions

Five new questions practising the same skills as the exam question above. They are not past exam questions. Marks are as in the original. Answers are at the end.

1 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $x\sqrt{3}+20 = x\sqrt{27}+2x$. Give your answer in the form $a\sqrt{3}+b$, where a and b are integers to be determined.

2 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $x\sqrt{7}+24 = x\sqrt{28}+3x$. Give your answer in the form $a\sqrt{7}+b$, where a and b are integers to be determined.

3 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $x\sqrt{3}+7 = x\sqrt{12}+2x$. Give your answer in the form $a\sqrt{3}+b$, where a and b are integers to be determined.

4 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $x\sqrt{5}+9 = x\sqrt{20}+2x$. Give your answer in the form $a\sqrt{5}+b$, where a and b are integers to be determined.

5 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $x\sqrt{2}+11 = x\sqrt{18}+3x$. Give your answer in the form $a\sqrt{2}+b$, where a and b are integers to be determined.

Question 1

$$x\sqrt{3}+20=3x\sqrt{3}+2x \;\Rightarrow\; 20=2x\sqrt{3}+2x \;\Rightarrow\; x=\frac{10}{1+\sqrt{3}}=\frac{10(\sqrt{3}-1)}{2}=5\sqrt{3}-5$$

Question 2

$$\sqrt{28}=2\sqrt{7}\text{, so } 24=x\sqrt{7}+3x=x(3+\sqrt{7}) \;\Rightarrow\; x=\frac{24(3-\sqrt{7})}{9-7}=36-12\sqrt{7}$$

Question 3

$$\sqrt{12}=2\sqrt{3}\text{, so } 7=x\sqrt{3}+2x=x(2+\sqrt{3}) \;\Rightarrow\; x=\frac{7(2-\sqrt{3})}{4-3}=14-7\sqrt{3}$$

Question 4

$$\sqrt{20}=2\sqrt{5}\text{, so } 9=x\sqrt{5}+2x=x(2+\sqrt{5}) \;\Rightarrow\; x=\frac{9(\sqrt{5}-2)}{5-4}=9\sqrt{5}-18$$

Question 5

$$\sqrt{18}=3\sqrt{2}\text{, so } 11=2x\sqrt{2}+3x=x(3+2\sqrt{2}) \;\Rightarrow\; x=\frac{11(3-2\sqrt{2})}{9-8}=33-22\sqrt{2}$$